We now peruse the relevant changes to the page.js component for the `handleOnSubmit` function together with some error-handling scenarios. The `handleOnSubmit` function demonstrates how to integrate the new structured data generation capability into the existing chat interface. It maintains the conversation flow while incorporating the new product list format for AI responses. The error handling ensures a graceful user experience even when problems arise during the AI interaction.

Listing 4.12 Updated page.js component form submit handler

```
const handleOnSubmit = async (event) => {
    event.preventDefault();
    const value = input.trim();
    setInput('');
    if (!value) return;
    setIsloading(true);
    setConversation((currentConversation) => [
      ...currentConversation,
      { id: generateId(), role: 'user', content: value },
    ]);

    try {
      const products = await
actions.generateProductList(value);
      const message = { role: 'assistant', products };
      setConversation((currentConversation) =>
[...currentConversation, message]);
    } catch (error) {
      console.error('Error generating product list:',
error);
      const errorMessage = { role: 'assistant',
products: [{ name: 'Error',
description: 'An error occurred while generating the
product list.', price: 0,
category: 'Error' }] };
      setConversation((currentConversation) =>
[...currentConversation, errorMessage]);
    } finally {
      setIsloading(false);
    }
  };
```

Calls the generateProductList server action with the user's input

Adds the AI-generated product list to the conversation state

Handles any errors that occur during product list generation

These changes lay the groundwork for a more dynamic and data-rich interaction between the user and our LLM. As we move forward to testing these changes, we'll be able to see how this new structured approach enhances the overall user experience of our application.

TESTING THE CHANGES

Now that we've implemented structured data generation in our Astra AI web application, it's time to put our changes to the test and explore the new capabilities we've unlocked.